Gaurika Mahajan

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Education

University of Toronto Toronto, ON
BASc in Computer Engineering, Minor in Artificial Intelligence *Sept 2022 – May 2027*
Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Networks, Databases, Computer Security, Deep Learning

Technical Skills

Languages and Frameworks: Java, Python, Golang, C++, C, Javascript, React, Express, TypeScript, HTML, CSS, MATLAB
Tools: Jenkins, Docker, Kubernetes, Terraform, Apache Kafka, Splunk, Datadog, SonarQube, TensorFlow, PyTorch
Databases/Cloud: AWS (EKS, S3, EC2, Lambda, DynamoDB), MongoDB, PostgreSQL, Redis

Work Experience

Software Engineering Intern | Zynga Toronto, ON | May 2025 – Present

- Migrated critical heartbeat API from legacy PHP endpoints to modern event-driven architecture, improving system reliability **by 40%** and reducing API response latency while maintaining backward compatibility for **10+ game titles**.
- Built centralized **CI/CD infrastructure** integrating SonarQube across **8+ repositories**, reducing code redundancy **by 65%**.
- Enhanced **Splunk logging** by building searchable dashboards and redesigning validation logic for **maintainability**.
- Led cross-team RPM migration via **Terraform** and **Jenkins**, improving system robustness and deployment success rate **by 28%**.

Full Stack Web Developer | UofT Engineering Orientation Website Remote | May 2024 – Aug 2024

- Developed and deployed **React/Redux** web application serving **1,000+ engineering students** using SDLC best practices, building **Express REST API** with real-time leaderboard functionality and performance-tracking data visualizations.
- Architected scalable backend infrastructure using **MongoDB** and optimized indexing to improve query performance **by 40%**.
- Implemented **Redis** caching for background subscription handling, enhancing system scalability **by 20%**.
- Integrated reliable payment flows via **Stripe API** and webhooks to process **$150,000+** in signup transactions.

Compliance Intern | Fresenius Kabi USA Chicago, IL | May 2024 – Aug 2024

- Designed monorepo architecture consolidating 7 repositories into single workflow, **improving data retrieval speed by 95%**.
- Developed data visualization pipeline, in **SmartSheet** and **MS SharePoint** for quarterly reports, speeding up tracking **by 80%**.
- Analyzed **100k+ datapoints across 12 years** to create database schema, track quality agreements, and extract insights that improved product launch forecasting **from 6 hours to seconds**.
- Automated call-to-action response alerts, **eliminating 90%** of manual tracking errors and enhancing operational stability.

Leadership

Software Development Director | UofT You're Next Career Network April 2024 – Present

- Led team of 10 developers in expanding **React/Firebase** platform serving **3,500+ students** to manage event sign-ups.
- Defined product roadmap, created technical specs for feature development, and designed user-centric interface in **Figma**.

Web Development Director | UofT Women in Science & Engineering Aug 2023 – Aug 2024

- Led website development and infrastructure improvements, enhancing user experience for **500+ members**.
- Architected automated email management system using **Gmail APIs**, reducing manual administrative overhead **by 60%**.

Projects

Code Canvas | Docker, LLM Integration, C4 Modeling, Claude Sonnet 4, Prompt Engineering

- Built AI-powered system architecture diagram generator via **Claude Sonnet 4** and C4 modeling to analyze repos **in minutes**.
- Engineered containerized solution capturing **85%** of codebase architecture, to produce system, infra, and component diagrams.
- Eliminated outdated documentation across **15+ engineering teams**, reducing onboarding time **by 60%**.

Skin Cancer Classifier | Python, PyTorch, TensorFlow, Google Colab, Git

- Developed a AI-driven CNN-based skin cancer classifier, achieving **80% test accuracy** on dermatoscopic images.
- Integrated unsupervised learning techniques for data augmentation and feature extraction to enhance training data quality.
- Led data collection, model creation, training, and hyperparameter optimization for improved classification performance.